Université Libre de Bruxelles

Lattice-based cryptanalysis of discrete-log signature schemes under partial nonce leakage

Student:
Ilias Elal-Lati
ULB ID: 000607880
Mohamed Amine Belmahi
ULB ID:000509057
David Botton
ULB ID: 000615056
Léandre Gerday
ULB ID: 000616822

Teachers:
Christophe Petit
Liran Lerman

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1 Project topic

Digital signature systems based on the discrete logarithm problem, such as the Digital Signature Algorithm (DSA) and the Elliptic Curve Digital Signature Algorithm (ECDSA), derive their security from the computational difficulty of solving discrete logarithms (Nguyen & Shparlinski, 2002; Boneh & Venkatesan, 1996). In these protocols, each signature requires an ephemeral secret value k that needs to be selected uniformly at random from $\mathbb{Z}_q^*$, with q representing the order of the relevant subgroup. Keeping both the secrecy and uncertainty of these nonce values is important, as any exposure or bias in their selection can enable adversaries to reconstruct the long-term private key α.

Nguyen and Shparlinski established that when an attacker possesses even partial information about the nonces across a sufficient set of signatures, it becomes possible to recover the private key in polynomial time.

The effects of these attacks in real life environments has been demonstrated by later research. Breitner and Heninger, for instance, used lattice-based analysis on publicly available ECDSA signatures obtained from major blockchains (such as Bitcoin, Ethereum, and Ripple) and from Internet-wide scans (including SSH and HTTPS). They found that biases in nonce generation can allow attackers to recover private keys.

This project investigates how lattice-based cryptanalysis can break digital signature schemes through such weaknesses. It includes hands-on experiments that replicate simplified versions of real attacks using tools like SageMath and assesses both the strengths and limitations of these methods.

2 References

Our initial references were:

- P. Q. Nguyen and I. E. Shparlinski, The Insecurity of the Digital Signature Algorithm with Partially Known Nonces, Journal of Cryptology, 2002.
- P. Q. Nguyen and I. E. Shparlinski, The Insecurity of the Elliptic Curve Digital Signature Algorithm with Partially Known Nonces, Designs, Codes and Cryptography, 2003.
- D. Boneh and R. Venkatesan, Hardness of Computing the Most Significant Bits of Secret Keys, CRYPTO 1996.

To better understand the historical development of the attack, we also added the following foundational reference:

- N. A. Howgrave-Graham and N. P. Smart, Lattice Attacks on Digital Signature Schemes, Designs, Codes and Cryptography, 2001.

Following the feedback on our proposal, we also added a more recent reference:

- J. Breitner and N. Heninger, Biased Nonce Sense: Lattice Attacks against Weak ECDSA Signatures in Cryptocurrencies, 2019.

This additional paper is particularly relevant because it demonstrates that attacks related to our topic are not merely theoretical.

3 Progress so far

So far, we have implemented a complete experimental reproduction of the lattice-based attack on ECDSA with partially known nonces.

We have developed a Python/SageMath codebase (available at `https://github.com/nottoBD/lattice-biased-nonce-ecdsa`) that generates synthetic ECDSA signatures on the secp256k1 curve with controlled least-significant-bit (LSB) nonce leakage. For each generated dataset, we verify that the known portion of the nonces matches the expected biased values.

We have implemented the computation of the hidden-number problem quantities t_i and u_i as defined in Nguyen and Shparlinski's paper. Finally, we have built and executed a Boneh–Venkatesan-style lattice attack using LLL reduction followed by Babai's nearest-plane algorithm.

Our implementation successfully recovers the correct private key in validated test cases, notably when 8 least significant bits of each nonce are known and a sufficient number of signatures is available.

4 Our understanding of the papers

We have studied in depth the two foundational papers by Nguyen and Shparlinski on lattice attacks against DSA and ECDSA with partially known nonces, the seminal work by Boneh and Venkatesan that introduced the Hidden Number Problem (HNP), and the more recent practical application by Breitner and Heninger.

The core technical contribution of Nguyen and Shparlinski begins from the standard ECDSA signature equations. Let $\alpha \in \mathbb{F}_q^*$ be the signer's secret key and G a base point of prime order q. For a message m, the signer chooses a random nonce $k \in \mathbb{F}_q^*$ and computes the signature (r, s) where

$$r = x(kG) \bmod q, \qquad s = k^{-1}\big(h(m) + \alpha r\big) \bmod q,$$

with $x(\cdot)$ denoting the x-coordinate of a point on the elliptic curve and h a cryptographic hash function. Rearrangement of the second equation immediately gives the linear relation

$$ks \equiv h(m) + \alpha r \pmod{q}.$$

Suppose that, for ℓ signatures, the attacker knows the t least significant bits of each nonce k_i. Then $k_i = k_i' + 2^t \tilde{k}_i$ where k_i' is known and $0 \le \tilde{k}_i < 2^{-t} q$. Substituting this expression and rearranging produces a collection of approximate modular linear relations

$$t_i \approx \alpha\, u_i \pmod{q},$$

where the quantities t_i and u_i are efficiently computable from the public values r_i, s_i, k_i' and $h(m_i)$, and the approximation error is bounded by $q/2^t$. This is precisely a variant of the Hidden Number Problem (HNP) introduced by Boneh and Venkatesan.

Nguyen and Shparlinski prove that such an ECDSA-HNP instance can be solved in polynomial time using lattice reduction. They construct a lattice of dimension $\ell + 1$ in which sufficiently short vectors correspond directly to the unknown private key α. The LLL algorithm followed by Babai's nearest-plane method (a polynomial-time estimation algorithm for the Closest Vector Problem) is then applied to recover a short enough vector from which α can be extracted. The authors show that the attack succeeds with overwhelming probability when $t \approx \frac{1}{2} \log_2 q$ and

$\ell = O(\log q)$, under standard assumptions on the hash function and the distribution of the nonces.

In our implementation (available at the GitHub repository linked in Section 3), we have reproduced this exact lattice construction on the secp256k1 curve. The code computes the HNP quantities t_i and u_i, builds the corresponding lattice, performs LLL reduction, and applies Babai's nearest-plane algorithm.

The implementation successfully recovers the correct private key α when only 8 least significant bits of each nonce are known and a sufficient number of signatures is supplied, which corresponds to the theoretical margins for small-to-medium security parameters.

5 Work distribution

6 Difficulties encountered

7 Next steps